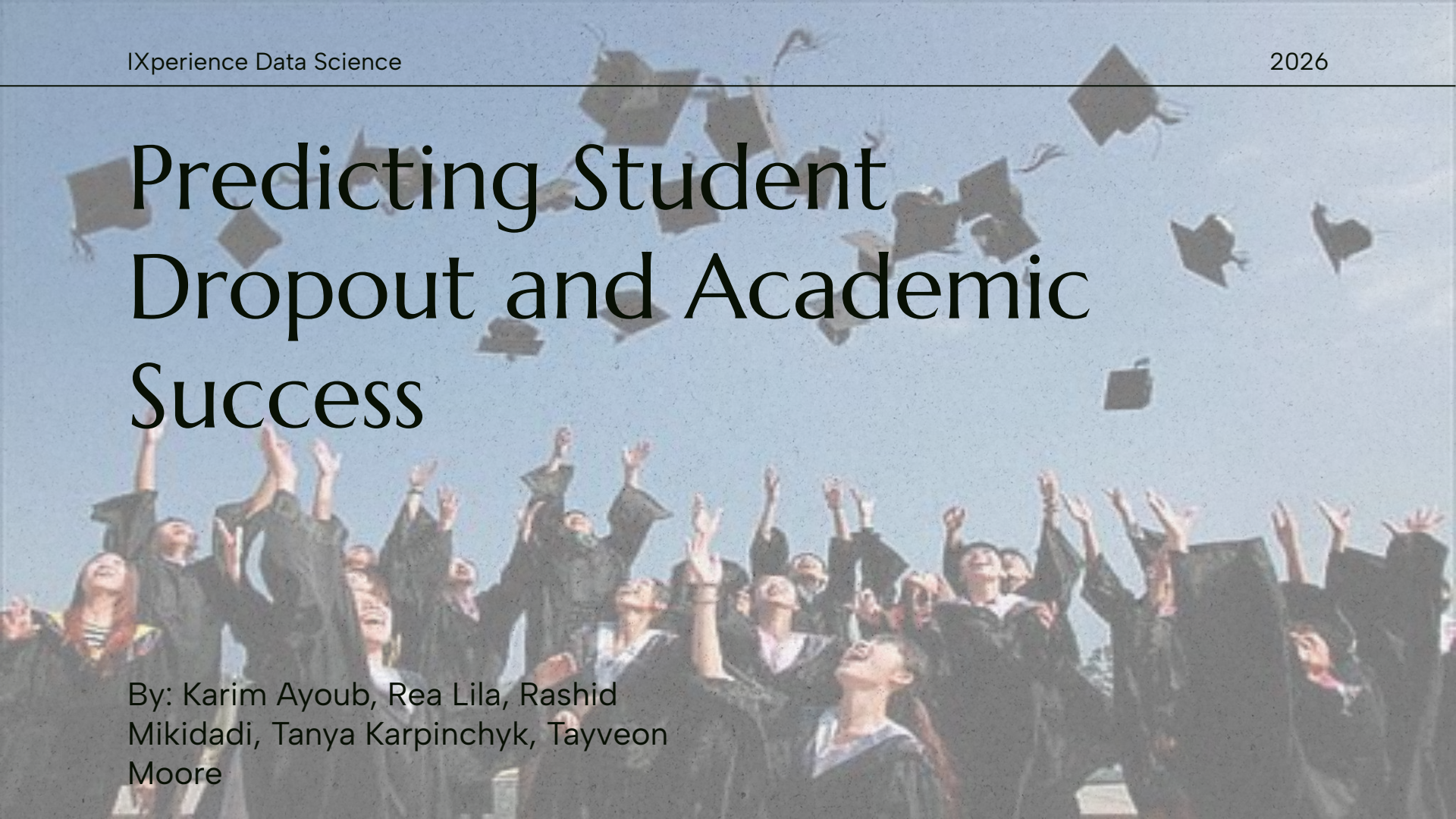
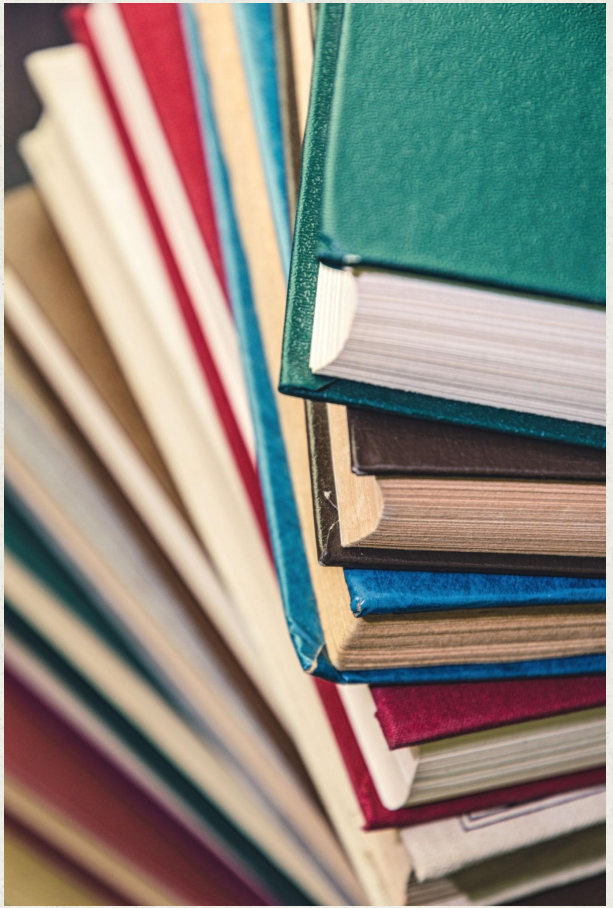


Predicting Student Dropout and Academic Success

By: Karim Ayoub, Rea Lila, Rashid Mikidadi, Tanya Karpinchyk, Tayveon Moore

Background

- Graduating from college allows opportunities valuable in academia, the workforce and life. (Higher pay, connections, etc)
- Identifying individuals who are likely to drop out can help with providing resources to foster the continuation of education which will lead to these opportunities
- Associating which features lead to removal from the college program can assist with greater work in approving barriers such as socioeconomic status and mental health when approaching higher learning



IXperience Data Science 2026

Goal

Objective:

- Create a predictive model in which institutions can identify students who may need support early enough to improve student outcomes

WHY:

- Earlier opportunities of support not only lead to academic success, they also increase graduation rates.



Findings



Highest correlation

The courses passed and student's financial status are the highest determining factor of if a student will graduate or dropout!



Early Action

To combat student dropout earlier Schools can look at student's prior grades/socioeconomic background to highlight campus resources



Next Steps

By identifying which students are at risk of dropping out, Schools can push students towards multiple scholarships or tutoring resources

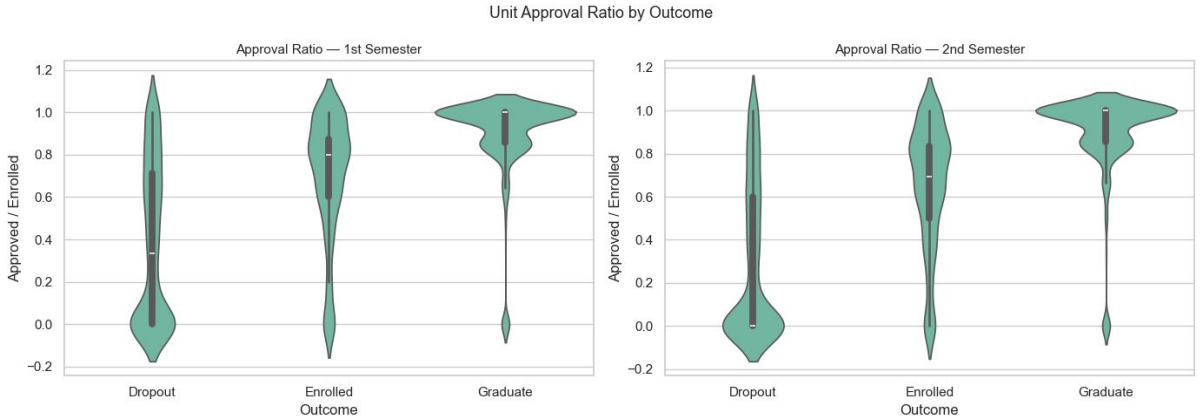
DATA PREPARATION

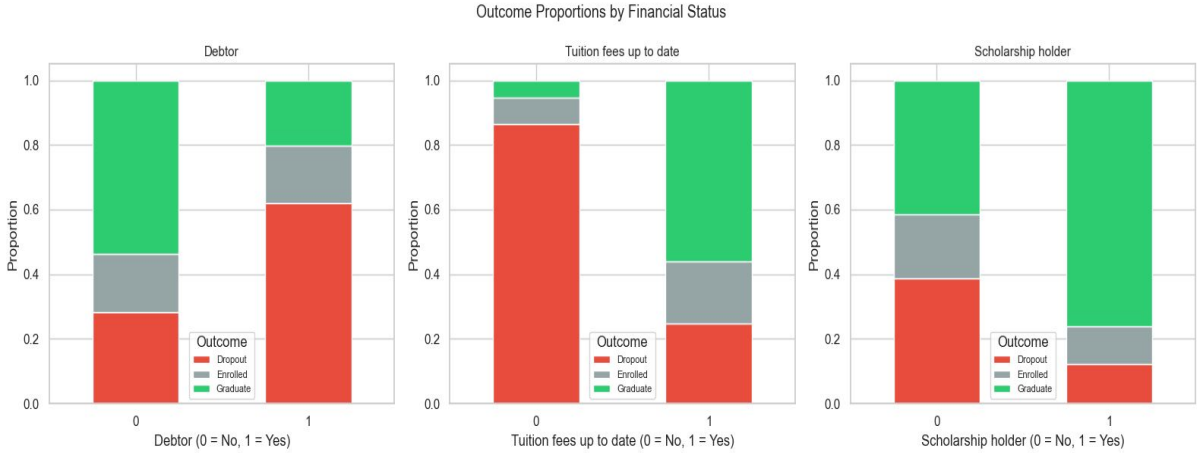
80/20 Train test split

Feature engineer – for all models:

- **financial risk flags (0–3)** representing ‘Debtor’, ‘Tuition fees up to date’, and ‘Scholarship holder’.
- **approval ratio (0–1)** for 1st and 2nd semesters.
    - what fraction of courses they actually complete (pass)

FEATURE IMPORTANCE - EDA







Modeling

01	Random Forest	→	Flagging at-risk students (at risk of dropping out vs not at risk)
02	XGBoost	→	3 class classification vs. Binary classification
03	Logistic Regression	→	Predicting whether a student will Graduate, Drop out, or stay Enrolled
04	SVM	→	Finding the optimal boundary to classify students as Graduate, Dropout, or Enrolled

RANDOM FOREST

01.

Random Forest Classifier

By creating many decision trees using individual features from the provided data, the model is able to aggregate a highly probable answer.

This model used an 80/20 split and resulted in binary predictions. The student was either **at risk** of dropping out or **not at risk** of dropping out.

```
Accuracy: 0.880225988700565

Confusion Matrix:
[[572  29]
 [ 77 207]]

Classification Report:
              precision    recall  f1-score   support

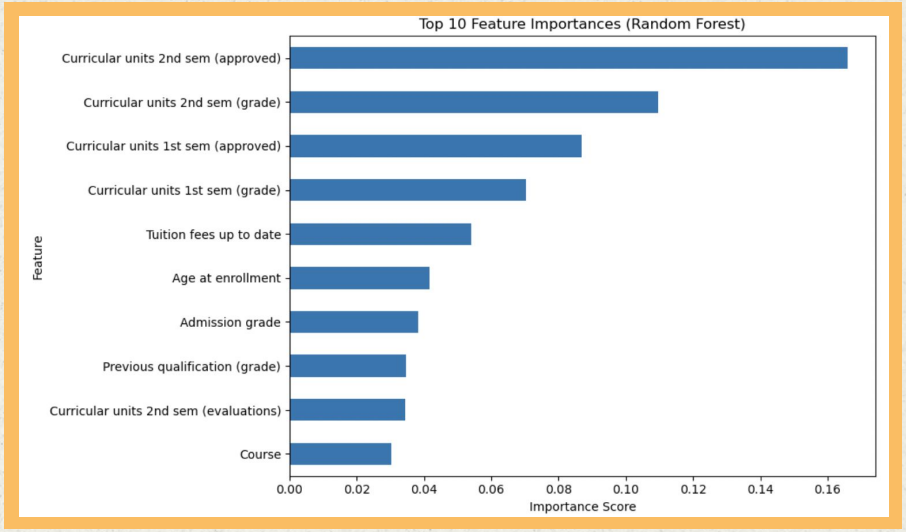
           0       0.88      0.95      0.92       601
           1       0.88      0.73      0.80       284

    accuracy                           0.88       885
   macro avg       0.88      0.84      0.86       885
weighted avg       0.88      0.88      0.88       885
```

Features & Takeaways

- Curricular units are consistently the most important feature
- Tuition fees, age, and grades follow closely behind

- Through using bagging, the model helps prevent overfitting
- Reduces variance by dividing data into smallest sets & testing parallel to each other (one outlier does not affect overall result)
- Achieving relatively high accuracy, precision, and recall, this model successfully helps identify at-risk students.



XGB CLASSIFIER

02.

3 Class Classification - XGBoost

WHY XGBoost

- Handles mixed feature types well
- Robust to multicollinearity
- No feature scaling needed

ENCODED TARGET

- 3 class classification: Target = 0 for dropouts, 1 for enrolled, 2 for graduates.

MODEL PARAMS

- multi:softprob + eval_metric='mlogloss'
- n_estimators=300, learning_rate=0.05
- max_depth=5

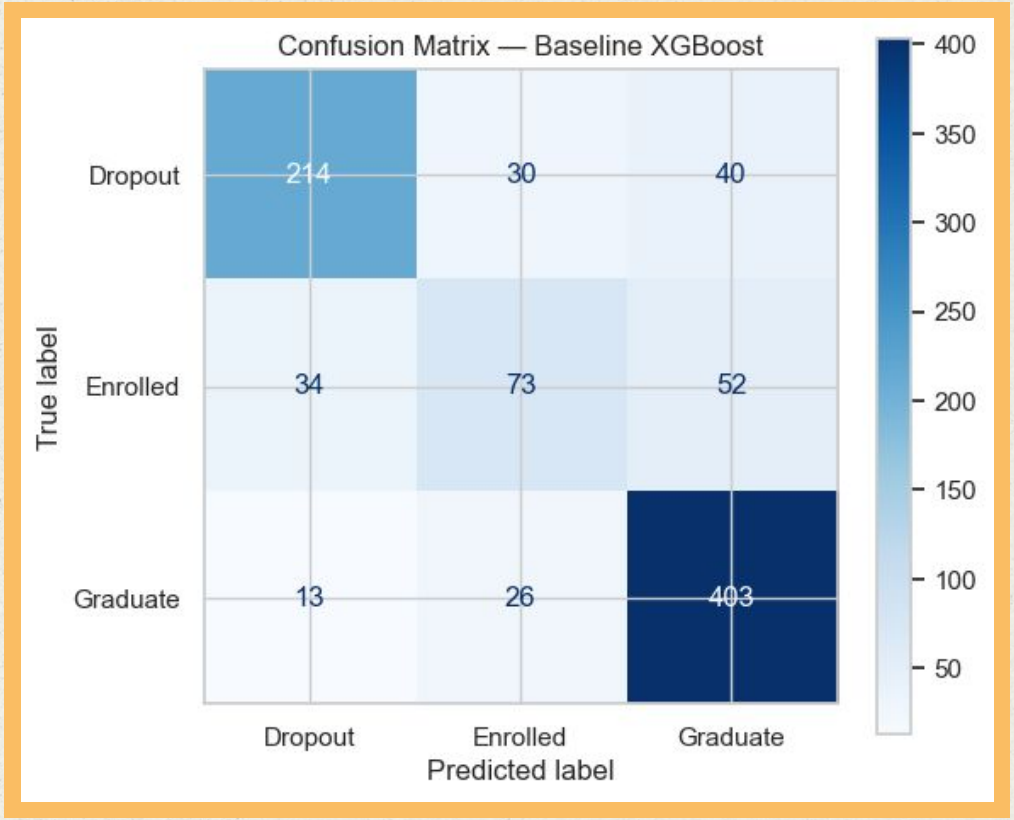
RESULTS

Macro F1: **0.717** — treats all three classes equally, so the minority Enrolled class actually counts

Graduate recall **0.91** — excellent

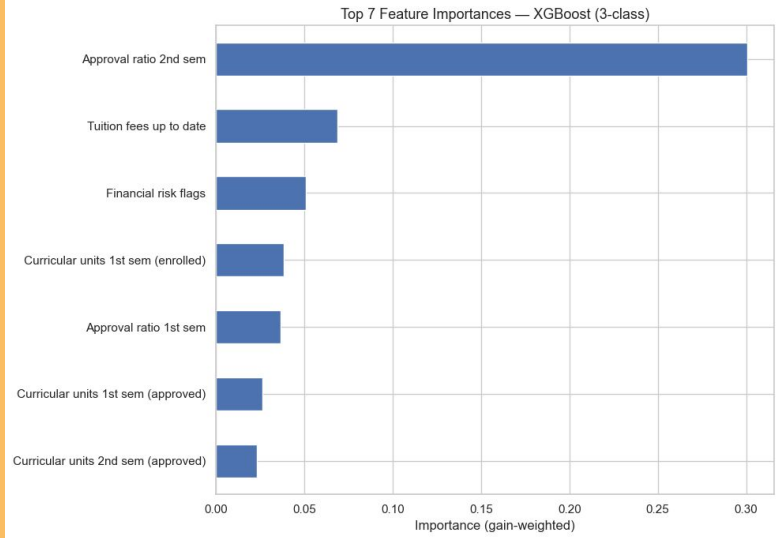
Dropout recall **0.75** — solid

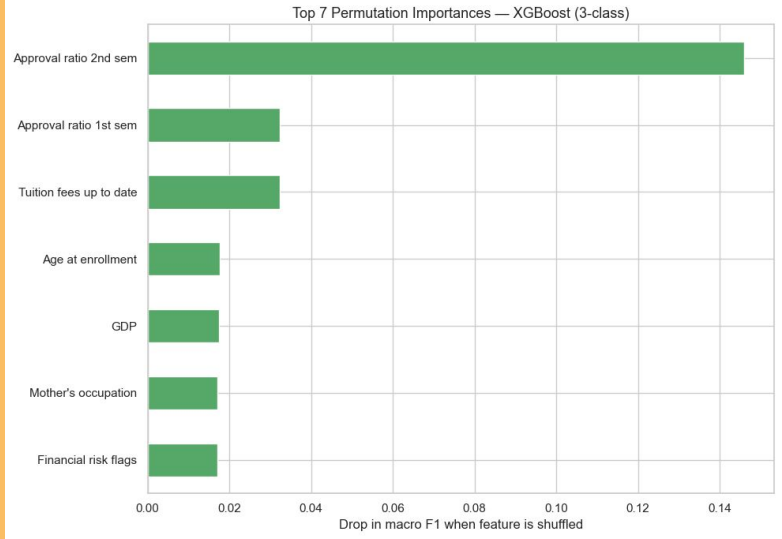
Enrolled recall **0.46** — low, but expected



FEATURE IMPORTANCE

- Approval ratio 2nd semester dominates
- EDA alignment
- Permutation test confirms it
- One notable difference, the permutation result is the more reliable one.





Binary Classification - XGBoost

APPROACH:

- Train on resolved students only (Graduate vs. Dropout)
- Apply the trained model to currently enrolled students

ENCODED TARGET:

- Binary classification: Target = 0 for graduates, 1 for dropouts

MODEL PARAMS:

- binary:logistic + eval_metric='logloss'
- n_estimators=300, learning_rate=0.05
- max_depth=5.

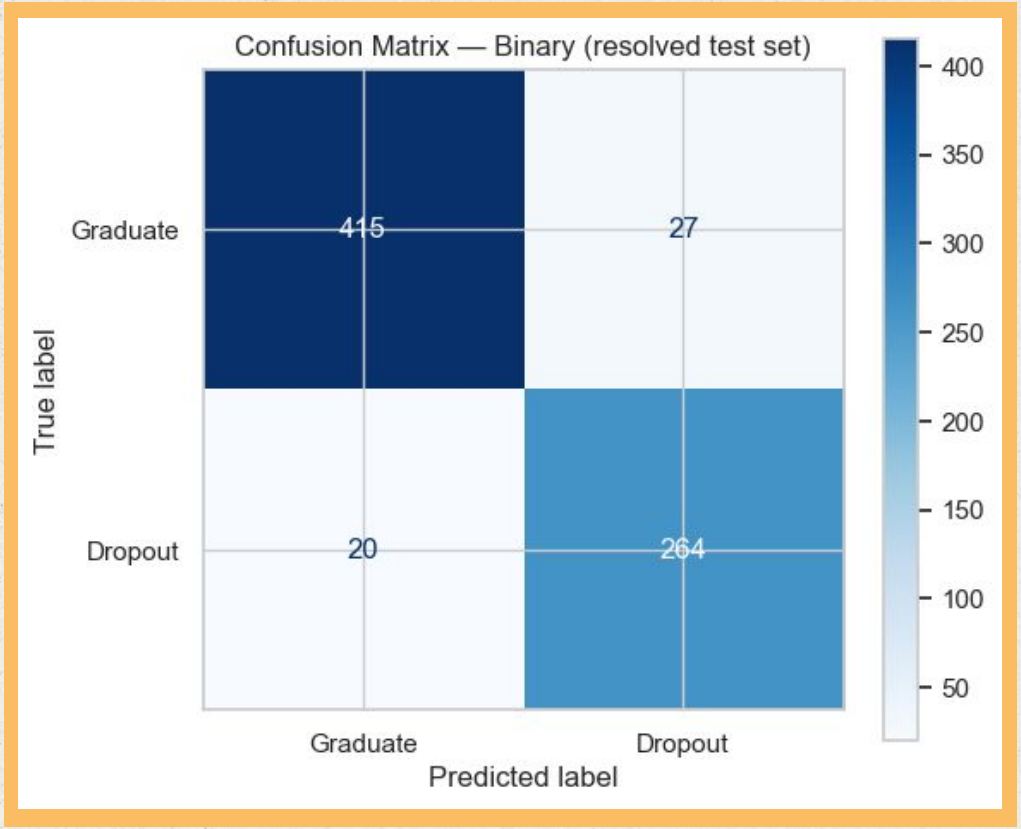
RESULTS

Macro F1: **0.93** — treats both classes equally

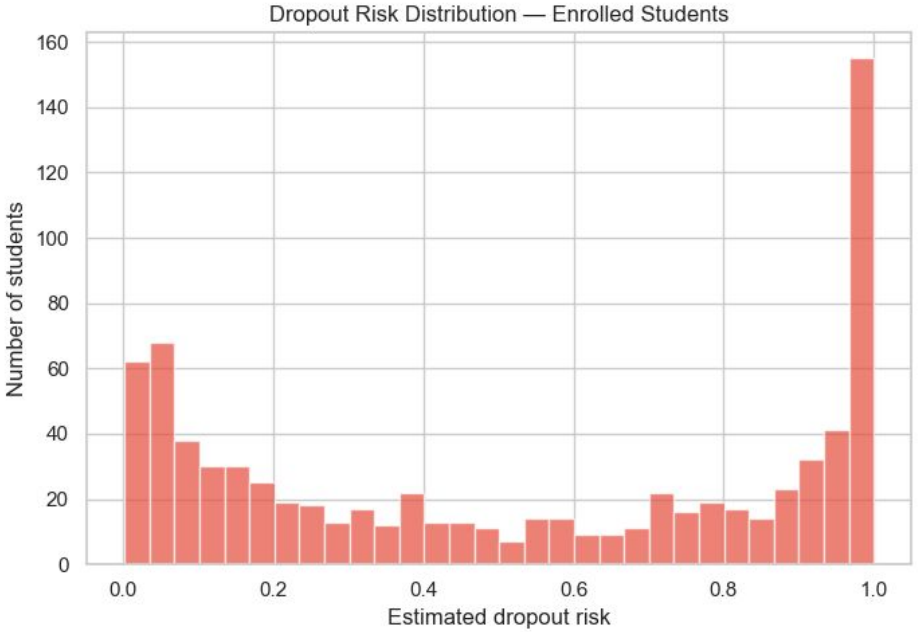
Graduate recall: **0.94** — excellent

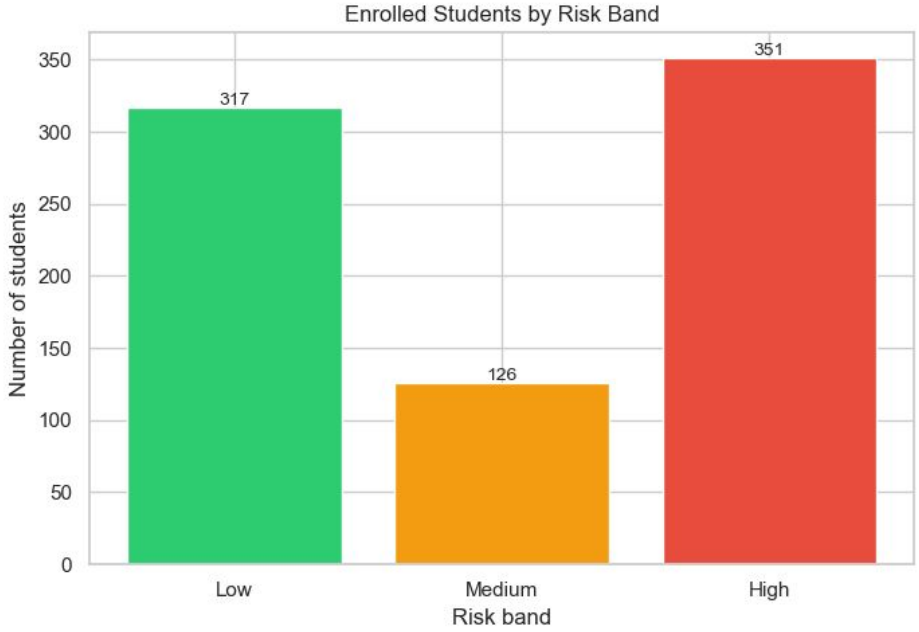
Dropout recall: **0.93** — catches 93% of at-risk students

ROC-AUC: **0.972** — near-perfect separation between graduates and dropouts



INTERPRETATION





LOGISTIC REGRESSION

Can we identify students at risk early enough to help them?

03.

4,424
students

3
outcomes

77%
accuracy

Logistic Regression

It is a simple, interpretable model that predicts which category something belongs to by weighing different factors.

I used it to predict each student's outcome:

- **Graduate** — Successfully finished
- **Dropout** — Left before finishing
- **Enrolled** — Still studying

Why Logistic Regression?

It's transparent.

We can see exactly ***why*** a student is flagged.

That matters when a school needs to justify reaching out to support a specific student.

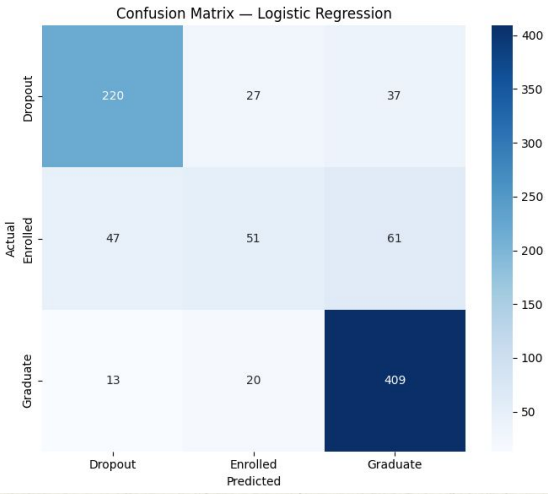
How well does the model work?

```
Classification Report:

              precision    recall  f1-score   support

     Dropout       0.79      0.77      0.78       284
    Enrolled       0.52      0.32      0.40       159
    Graduate       0.81      0.93      0.86       442

    accuracy                           0.77       885
   macro avg       0.70      0.67      0.68       885
weighted avg       0.75      0.77      0.75       885
```



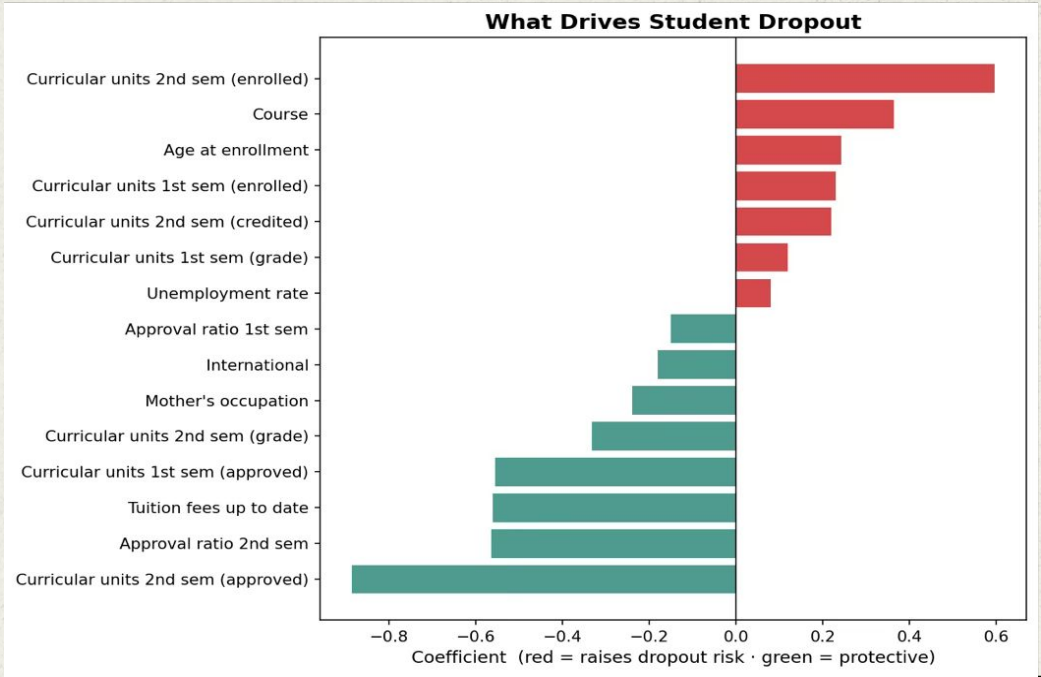
Per-class performance

Outcome	Recall	AUC
Graduate	0.93	0.92
Dropout	0.77	0.91
Enrolled	0.32	0.80

The key insight: The model catches 77% of real dropouts — about 3 in 4 at-risk students. The **Enrolled** group is hardest, because those students look like both future graduates and future dropouts.

What drives dropout?

Because the model is interpretable, we can read which factors raise or lower dropout risk



Raises dropout risk

Enrolling in many units, older age at enrollment, and certain courses push a student toward dropping out.

Protects against dropout

Passing units, a high approval ratio, and paying tuition on time strongly predict graduating.

The findings — and is the model good?

Putting the model to work

The model ranks every student by dropout probability, so the school gets a **ready list of the highest-risk students** to support first.

Bonus: the strongest signals come from **2nd-semester results** early enough in a program to step in and help.

My honest verdict

The model is decent at what it does

- ✓ Catches 3 of 4 dropouts (recall 0.77)
- ✓ Strong AUC (0.91 for dropout)
- ✓ Fully explainable

- ✗ Weak on the Enrolled middle group
- ✗ 77% accuracy leaves room to improve

The other models may score higher, but logistic regression gives us something valuable: a clear, honest starting point we can explain and act on.

Support Vector Machine (SVM)

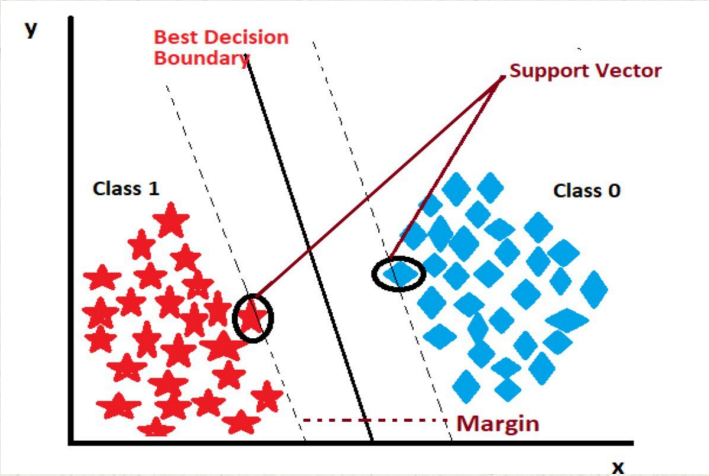
04.

Support Vector Machine (SVM)

Think of every student as a dot on a graph, positioned based on their grades, tuition status, and other features. SVM finds the widest possible gap between the groups of dots and draws a boundary line through it. The wider that gap, the more confident the model is when classifying a new student.

Three possible student outcomes

- Graduate
- Dropout
- Enrolled



Why SVM?

It finds boundaries, not rules.

- SVM takes a completely **different approach** by separating students geometrically, not using trees or probabilities
- Finds the **widest possible margin** between outcome groups, making predictions more confident and robust
- Performs well in **high dimensional space**, this dataset has 38 features
- Works well when **classes** overlap, which is expected with Graduate, Dropout and Enrolled students
- **Effective** with scaled data and handles class imbalance well with balanced weights

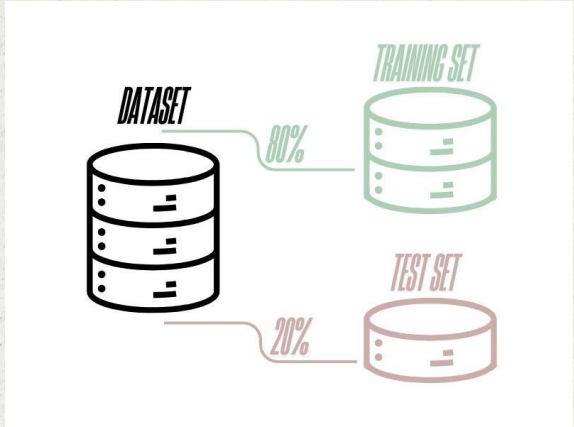
Model Setup

How was the model built?

- **RBF kernel** was chosen because student data has complex non-linear patterns that a straight line cannot capture
- **StandardScaler** was applied because SVM is very sensitive to feature scale — without it, features like age and grades would unfairly dominate
- Stratified split ensures the same proportion of Graduate, Dropout and Enrolled students appear in both training and test sets

Model Setup

- Kernel: **RBF** to capture non-linear patterns
- class_weight="balanced" to handle class imbalance
- Features standardized using **StandardScaler**
- **80/20** stratified train/test split
- probability=True to enable ROC/AUC curves



Results

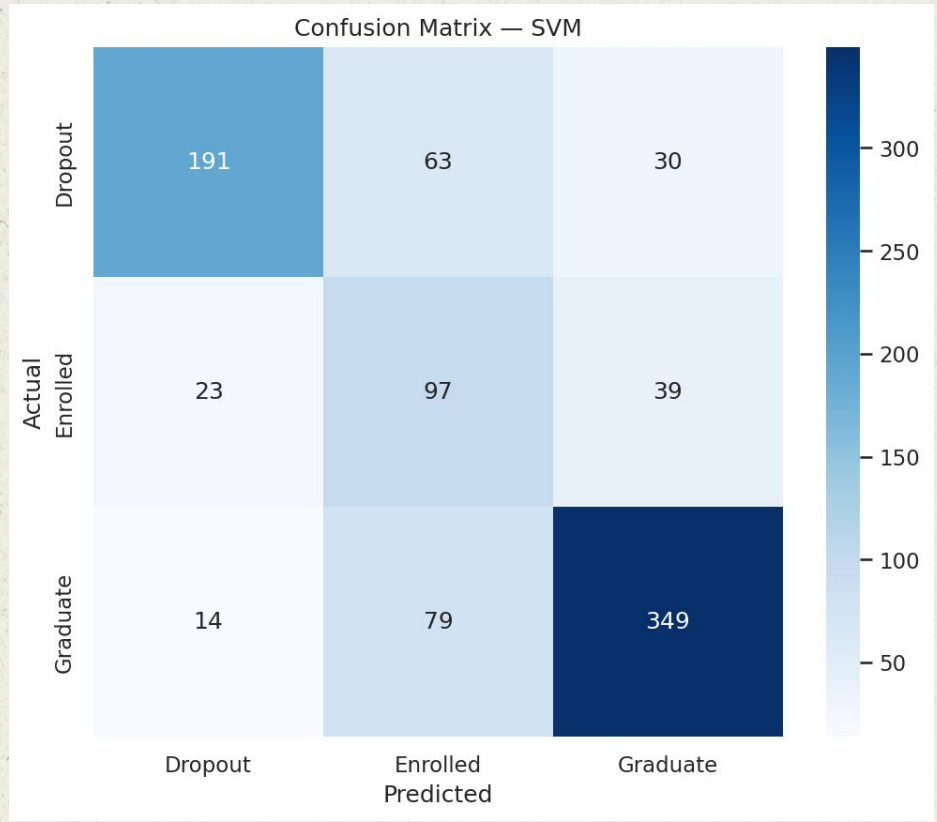
How did the model perform?

- Overall accuracy: **72%**
- Graduate F1: **0.81** — predicted best
- Dropout recall: **0.67** — correctly identifies 2 in 3 at-risk students
- Enrolled F1: **0.49** — hardest class, expected given its ambiguity
- Per-class metrics **matter** more than accuracy here because the classes are imbalanced

SVM Accuracy: 0.7198

Table +

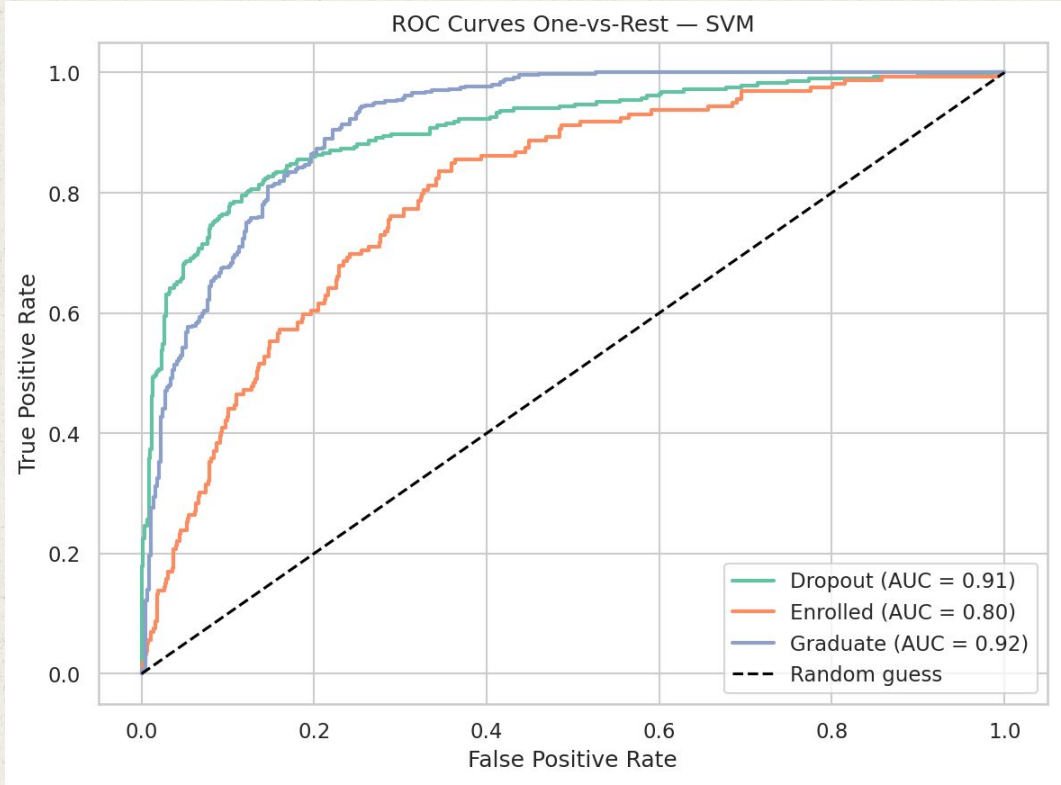
	class	precision	recall	f1-score	support
1	Dropout	0.838	0.673	0.746	284
2	Enrolled	0.406	0.61	0.487	159
3	Graduate	0.835	0.79	0.812	442
4	macro avg	0.693	0.691	0.682	885
5	weighted avg	0.759	0.72	0.732	885



ROC/AUC Results

How well does it rank students by risk?

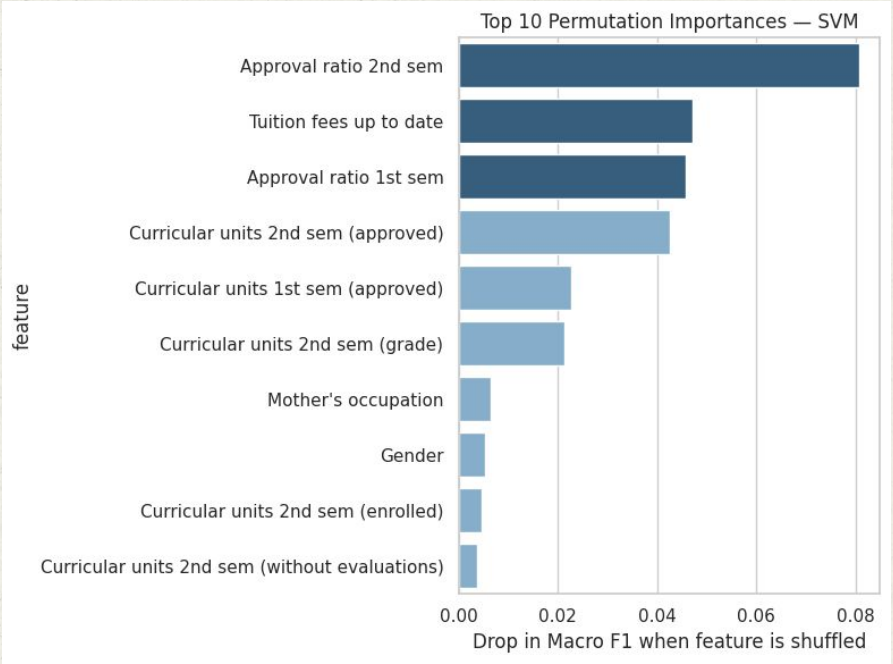
- Graduate AUC: **0.92** — outstanding separation
- Dropout AUC: **0.91** — strongly ranks at-risk students
- Enrolled AUC: **0.80** — weakest but still excellent
- All curves sit well above the random guess diagonal
- A Dropout AUC of 0.91 means an institution can reliably rank at-risk students and intervene before it is too late



Feature Importance and Takeaways

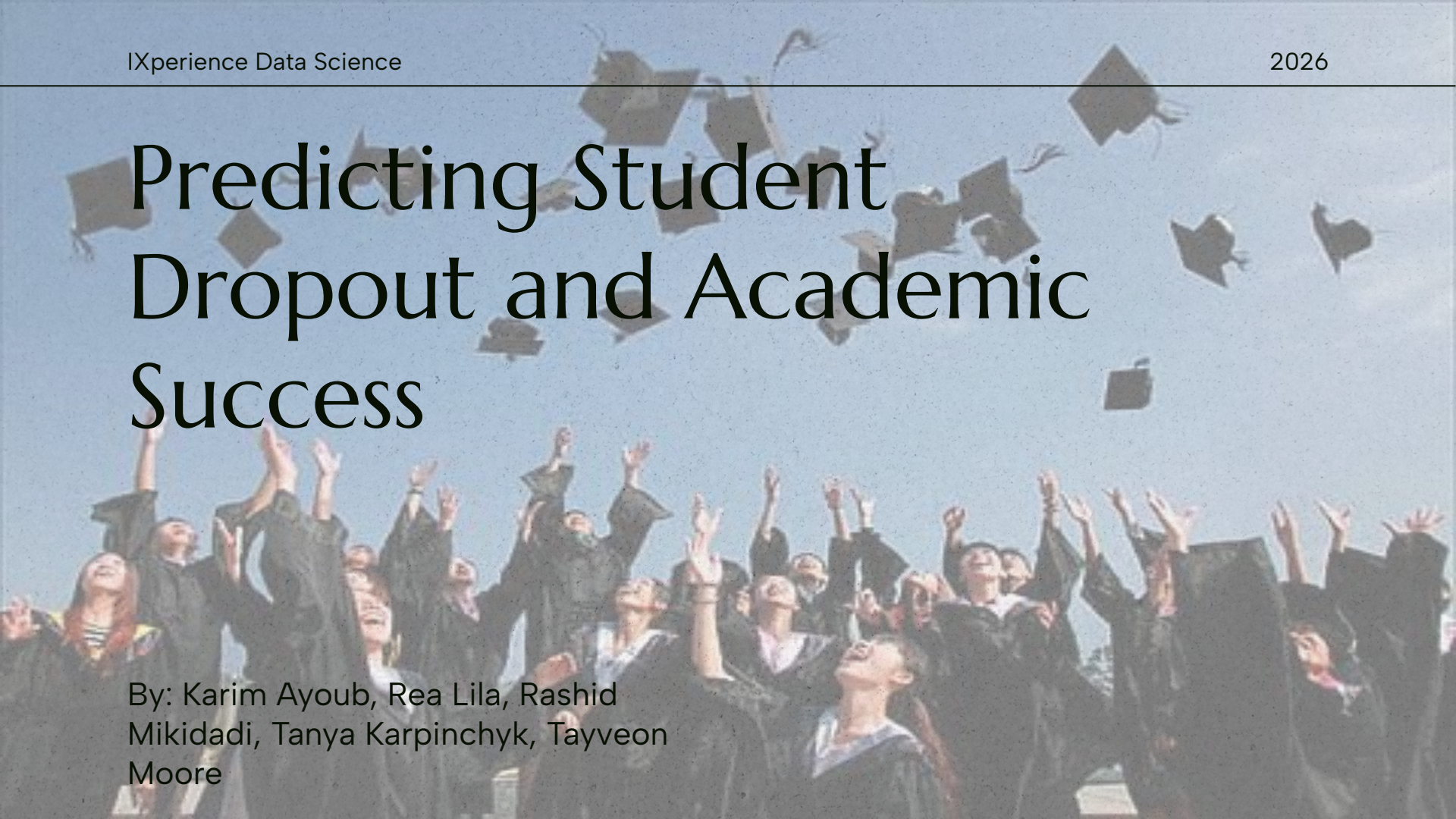
What drives the SVM predictions?

- **Approval ratio 2nd sem** is the strongest signal by far
- **Tuition fees up to date** is second
- **Approval ratio 1st sem** is third



SVM Key Takeaways

- The model catches **2 in 3 students** who actually drop out
- Dropout AUC of **0.91** means it reliably ranks at-risk students for early intervention
- Graduate and Dropout are separated very well despite using no tree based rules
- Enrolled is genuinely the hardest outcome to predict, not a weakness of SVM specifically
- The RBF kernel allowed SVM to capture complex patterns that a linear model would miss



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